

Practice Session 2

Part 1: Measures of central tendency for quantitative data

1.1 Calculating the Sample Standard Deviation by Hand

Here is the formula for the sample mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

And here is the formula for the sample standard deviation:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Using the following data, complete the table for calculating the sample standard deviation.

Cost(Dollars)	b. Deviations: $x_i - \bar{x}$	c. Deviations squared: $(x_i - \bar{x})^2$
850		
900		
1400		
1200		
1050		
750		
1250		
1050		
565		
1000		

d. Write the sum of squared deviations value here:

$$\sum_{i=1}^n (x_i - \bar{x})^2 =$$

e. Write the value of the squared deviations divided by $n - 1$ here:

$$\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} =$$

f. Write the value for the square root of the sum of squared deviations here:

$$s = \sqrt{\quad} =$$

1.2 Five-Number Summary

Using the numbers from the previous exercise, do the following:

- Find the 5-number summary (minimum, Q1, median, Q3, maximum)
- Check your work using R functions

Answers

```
# Your code here

## you can use function fivenum()
```

1.3 Boxplots and Histograms

Consider the `mtcars` data set. This dataset is built into R, so you can access it directly; no downloads required!

You can learn more about this dataset using `?` in the console or the `View()` function. You can also use the functions `head()` and `tail()`.

1. First, create a histogram of the variable `mpg`. Then create a boxplot of `mpg`.

2. How do these two plots compare?
3. Create a boxplot of `mpg` by number of cylinders `cyl`.

Answer

```
#1# Create a histogram of 'mpg'

# Your code here


#2# Create a boxplot of 'mpg':

# Your code here


#3# create boxplot of mpg per cylinder:

# Your code here
```

1.4 Quantitative data : histograms and outliers

Generate histograms for each of the following data sets. Use the `$` command to access the individual data sets.

1. For each histogram, add the mean to the plot using `abline()`.
2. Add the median using `abline()`.
3. Do you see any potential outliers? Also calculate the five-number summary for each using R.

```
set.seed(999)
data = data.frame(
  dat1 = -rchisq(1000, df = 1),
  dat2 = rchisq(1000, df = 1),
  dat3 = runif(1000),
  dat4 = rnorm(1000),
  dat5 = sample(c(rnorm(1000, mean = 2), rnorm(1000, mean = 10)), size = 1000)
)
```

Answers:

```
# Your code here

#text(x = median1, y = 600, cex= 0.5, labels = "median",
      #adj = 0.5, col = "red") # Add label slightly above the line

#text(x = mean1, y = 650, cex= 0.5, labels = "mean",
      #adj = 0.5, col = "blue") # Add label slightly above the line
```

1.5 Percentiles

Compute the 25th, 50th, and 75th percentiles for the 5 datasets in the `data` data.frame. Which has the smallest median? Which has the largest? You can use the function `quantile()`.

```
# percentiles of data 1 percentiles
## Your code here

# percentiles of data 2
## Your code here

# percentiles of data 3
## Your code here

# percentiles of data 4
## Your code here

# percentiles of data 5
## Your code here
```

Part 2: Normal Distribution and +/- 2 Standard Deviations

The normal distribution (also known as the “bell-curve”) occurs very frequently in mathematics, statistics, and the natural and social sciences. Which of the 5 data sets in the `data` `data.frame` appears to be normally distributed?

2.1 Calculate the endpoints of one standard deviation above and below the mean.

Using this data set, find the mean and standard deviation, then calculate the endpoints of **1 standard deviation above**, and **1 standard deviation below the mean**. What percentiles do these values correspond to?

Note: Approximately 68% of the data lie between 1 standard deviation above and 1 standard deviation below the mean.

```
# Your code here
```

- we will use `dat4`

```
# Your code here
```

2.2 z-Scores

Expressing a distance from the mean in standard deviations standardizes the performance. To **standardize** a value, we subtract the mean and then divide this difference by the standard deviation:

$$z = \frac{y - \bar{y}}{s}$$

The values are called **standardized values**, and are commonly denoted with the letter z . Usually we just call them **z-scores**.

z -scores measure the distance of a value from the mean in standard deviations. A z -score of 2 says that a data value is two standard deviations above the mean. It doesn't matter whether the original variable was measured in fathoms, dollars, or carats; those units don't apply to z -scores.

Data values below the mean have negative z -scores, so a z -score of -1.6 means that the data value is 1.6 standard deviations below the mean.

Of course, regardless of the units, the farther a z -score is from 0, the more unusual it is. Thus, a z -score of -1.3 is more extraordinary than a z -score of 1.2 .

2.3 Example of comparing z -Scores

A town's January high temperatures average 2°C with a standard deviation of 6° , while in July the mean high temperature is 24°C and the standard deviation is 5° .

In which month is it more unusual to have a day with a high temperature of 13°C ? Explain.

Answers:

```
# January z-score
## Your code here

# July z-score
## Your code here
```

Extra practice: The Relationship Between Two Quantitative Variables/ Correlation and Regression

In this session you might use the formula of the correlation between two quantitative variables:

$$r_{xy} = \frac{1}{(n-1)s_x s_y} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Remember that the fitted regression line is defined by the equation:

- $\hat{y} = a + bx$, or
- $\widehat{\text{Response}} = a + b \cdot (\text{Explanatory})$

- Residuals = observed - predicted = $y - \hat{y}$

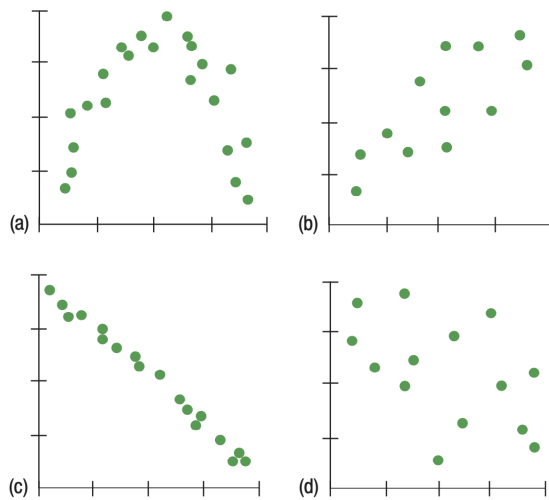
Where:

- Response: is the response variable or the dependent variable
- Explanatory: is the independent variable
- a: is the y-intercept
- b: is the slope of the regression line

You may use the following R functions: `plot()`, `lm()`, `cor()`, `abline()`. And you might need to download Lock5Data using `library(Lock5Data)`.

3.1 Describe scatterplots and correlation

Here are several scatterplots. The calculated correlations are 0.006, -0.977, -0.487, and 0.777. Match each scatter plot with the appropriate correlation coefficient.



3.2 Create scatterplots in R

Load the data `FloridaLakes` from `library(Lock5Data)`.

1. Describe the type of each of the variables `pH`, `Calcium`, and `Alkalinity`.
2. Create three scatterplots for each pair of the variables: `pH` vs `Calcium`, `pH` vs `Alkalinity`, and `Calcium` vs `Alkalinity`. Add the main title to each plot.
3. What is the correlation coefficient between `pH` and `Calcium`? Is it positive or negative?
4. What do these coefficients mean in the context of this data?
5. Try to calculate the correlation coefficient between `pH` and `Calcium` without using an R function.

```
# download the data and load it into R
library(Lock5Data)
data(FloridaLakes)
```

1. Describe the type of each of the variables `pH` and `Calcium`, `Alkalinity`.

The variables `pH`, `Calcium`, and `Alkalinity` are quantitative (continuous) variables.

2. Create scatterplots for each pair of variables (add a main title to each plot).

```
## Your code here
```

3. What is the correlation coefficient between `pH` and `Calcium`. Is it positive or negative.

```
## Your code here
```

4. What do these coefficients mean in the context of this data?

A correlation coefficient of 0.577 between `pH` and `Calcium` means that there is a linear positive association of moderate strength of 0.577.

5. Calculate the correlation coefficient between `pH` and `Calcium` without using an R function.

```
## Your code here
```